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Error Pattern Analysis: Negation Scope Detection with CRF

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Error Pattern Analysis Report

Introduction

In this report, a systematic error analysis is performed based on the results of the CRF classifier for the task of negation scope detection given the cue. First, errors are classified into error patterns, and then a counterfactual analysis is conducted to assess whether these error patterns represent genuine issues that affect classification.

Error Sample Selection

An error sample was selected from the combined predictions file of both test and development data by collecting a total of fifty random tokens with either false positive or false negative predictions. Subsequently, twenty-one¹ error categories were then identified both in respect to the 'Annotation of negation cues and their scope Guidelines' and the Universal Dependency Relations, four of which are the most prevalent in the error sample and are presented below.

Error Patterns

MC_and_SC:

Frequency: 6

This category includes sentences where the negation cue is in the main clause, the root is in scope, and the sentence contains a subordinate clause which is also in scope. According to the guidelines, if the main clause semantically affects the subordinate clause, then the subordinate clause is in scope, a rule that the CRF model may fail to capture due to its semantic base. This category includes the following errors:

1. Sentences where the subordinate clause (ccomp, relcl, advcl) is missing in the predictions
Prediction: [There 's] **no** [link between the two save for Henderson 's own servant], who serves the family 's meals.
Gold: [There 's] **no** [link between the two save for Henderson 's own servant, who serves the family 's meals].
2. Sentences where the subordinate clause depends on a different subordinate clause, and the negation is in the latter
Prediction: I was sure Henderson, as he calls himself, felt that he was suspected, and that [he would lie low and make] **no** [move so long as he thought he] was in any danger.
Gold: I was sure Henderson, as he calls himself, felt that he was suspected, and that [he would] lie low and [make] **no** [move so long as he thought he was in any danger].

¹ The complete list of error patterns can be found in the appendix.

Sub_CL_only:

Frequency: 5

This pattern includes sentences with a negation that only affects the subordinate clause it is in and not the clause it depends on (either main or another subordinate clause).

Prediction: Telegraphic inquiries had been already made [which showed that Marx knew] **nothing** [of his customer save that he was a good payer].

Gold: Telegraphic inquiries had been already made which showed that [Marx knew] **nothing** [of his customer save that he was a good payer].

Pred_exp_OOS:

Frequency: 5

This category refers to sentences where a predicative expression that follows a copulative verb is marked as out of scope by the CRF classifier.

Prediction: [Your cipher was] **not** difficult, madam.

Gold: [Your cipher was] **not** [difficult], madam.

Aff_skipped:

Frequency: 4

In cases where the negation cue is an affix attached to an adjective or adverb, the scope is marked as empty. The CRF classifier fails to recognize the adjective or adverb connected to the negation cue in both of the following situations:

1. When the adjective refers to the subject, and only the subject is within the scope.
2. When the adjective refers to something other than the subject, causing the entire clause to be in scope.

For adverbial negation cues, the scope extends to cover the entire clause.

Prediction: “Remarkable, but by no means **im**possible,” said Holmes, smiling.

Gold: “Remarkable, but by no means **im**[possible],” said Holmes, smiling.

Counterfactual Analysis on Two Error Patterns

Relevant cases for these error patterns were identified based on their dependency structures.

1. MC_and_SC

To search for this error pattern throughout the dataset, the spaCy dependency labels *ccomp*, *relcl*, and *advcl* were used to identify the subordinate clauses. For this pattern to be applicable, certain conditions needed to be met; the negation cue is in the main clause, the root is in scope, and the sentence contains a subordinate clause which is also in scope. After examining the results, another condition was added to locate whether the prediction did not include the root of the sentence or the subordinate clause. It is important to note that, in order to determine

whether a sentence was correct or incorrect, only the status of *ROOT* and *ccomp*, *relcl*, or *advcl* were assessed. An example of the findings can be found below:

Prediction: [We have] **no** [excuse for an intrusion upon his privacy until we have some reason] to think that there is a guilty reason for it.

Gold: [We have] **no** [excuse for an intrusion upon his privacy until we have some reason to think that there is a guilty reason for it].

The results were as follows:

Number of correct sentences: 36

Number of incorrect sentences: 19

Number of sentences where the root is missing in the predictions column: 6

Number of sentences where the subordinate is missing in the predictions column: 13

Out of the fifty-five sentences affected by this phenomenon, about 34% did not have the correct scope. Subsequently, this negation scope pattern is impactful enough to be considered significant. As it also involves semantic relations that are difficult to capture by the model, capturing whether a subordinate clause is directly affected by the meaning of the main clause or not is something that even human annotators can struggle with. Additionally, several of our features, such as token distance, dependency path, and dependency path length, may have confused the model, as the distance between the negation cue in the main clause and the subordinate clause could be quite large.

2. *Pred_exp_OOS*

This pattern was located with the use of *acomp* to determine whether an adjective following a copulative verb is in scope, but not in the predictions column of the file. The conditions for this rule to apply ensure that the word of the scope column is in the word of the predictions column to account for adjectives that contain a negation cue themselves. An example of the findings is presented below:

Prediction: “[Your case is] **not** complete, then?” I asked.

Gold: “[Your case is] **not** [complete], then?” I asked.

The results were as follows:

Number of sentences with *acomp* in scope and scope in pred: 44

Number of sentences with incorrect *acomp* predictions: 9

Out of the fifty-three sentences which were identified in the analysis, about 16% was handled incorrectly by the CRF classifier. While the percentage is not too great, it is an area that the classifier could improve in.

Conclusion

Accounting for an error margin in this analysis, the results are promising. In all cases analysed, the status of the necessary components (e.g. *ROOT*, *acomp*, *ccomp*, etc.) was checked to

determine whether the scope of a sentence was predicted correctly in accordance with the phenomenon investigated.

To conclude, even though the CFR classifier needs improvement to accurately handle complex sentence structures involving multiple subordinate clauses, the outcome is generally positive. While the classifier's overall predictions are highly accurate, improvements in semantic understanding and handling of such syntactic structures are necessary for better scope prediction.

References

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Appendix: Additional Error Categories

The following error categories were identified but not described in detail within the main report.

1. SC_not_MC: secondary clause in scope, but classifier also predicted main clause in scope
2. Wrong_subj: two subjects in a sentence, only one affected by the negation, but both classified as in scope, or two subjects in scope but only one marked
3. Prep_to_C: preposition only affects noun phrase but classifier predicted entire clause as in scope
4. Repeat_OOS: Repeated words confused as in scope when the negation cue examined is in fact in the other clause of the sentence
5. Aff_skipped: affix negation entirely skipped
6. No_pattern: random word in/out of scope without apparent syntactical or other reason
7. MC_not_SC: If the main clause semantically affects subordinate clause, then the subordinate clause is in scope. In this error category it is not in scope.
8. Sub_CL_only: the negation only affects the subordinate clause it is in and not the clause it depends on (either main or other subordinate)
9. PP_OOS: Prepositional phrase in the beginning of a negated sentence marked as in scope
10. Adv_OOS: Adverb not in scope, could be considered in scope since it is in the same clause as the negation depending on how scope is considered
11. MC_and_SC: If the main clause semantically affects subordinate clause, then the subordinate clause is in scope. In this category, the subordinate clause is affected. (includes cases where the model has failed to predict the root but predicted the subordinate clause)
12. Int_NIS: Interjection not in scope
13. NP_only: Infixal negation in noun phrases affects only the noun
14. MN_WC: Multi-word negation affects the entire clause, not just the NP
15. Diff_pred: One predicate of a main clause does not affect another predicate of that clause
16. Other_CL: One coordinate clause with negation does not affect another\
17. Adv_v: If the negated affix is attached to an adverb that is a complement of a verb, the negation scopes over the entire clause. This case was handled correctly by our model according to the Guidelines.
18. Adj_to_clause: Is an adjective negated with an affix? The negation scopes over all the clause if the adjective does not modify the subject

19. Pred_exp_OOS: Predicative expression marked as out of scope
20. Pro_aff_miss: pronoun missed in affix negation
21. PP_skipped: prepositional phrase skipped at the end of sentence